Ibn Tofail University

Rigid body mechanics

Problem Set IV

Exercise 1:

A perfect rigid body having the shape of a square $ABCD$ is moved from position $A_1B_1C_1D_1$ to position $A_2B_2C_2D_2$.

1. Show, using a figure, that the motion of this solid can be considered as the sum of a translation and a rotation about a suitably chosen fixed point.
2. Give the value of the angle of rotation.

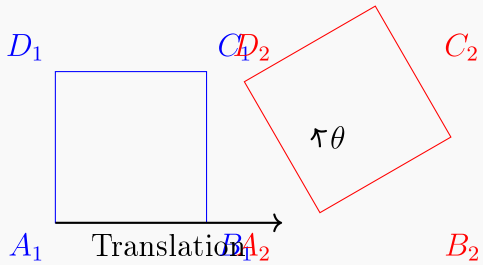
Correction

1. Motion as Translation + Rotation:
Any motion of a rigid body can be described as:

- A **translation** of the center of mass (or any point)
- A **rotation** about an axis passing through that point

For the square, we can choose point A (or the center O of the square) and show that:

- Translate point A from A_1 to A_2
- Rotate the square about A by angle θ



2. Angle of Rotation:
The angle of rotation θ depends on the specific positions. For example:

- If the square rotates about its center by a quarter turn: $\theta = 90° = \frac{\pi}{2}$ rad
- If it rotates about a vertex, θ is the angle between the initial and final positions of a side

In general: θ is the angle between any line fixed in the body in its initial and final positions.

Exercise 2:

Let $R(O, X, Y, Z)$ be a reference frame assumed fixed. A homogeneous rod of length L and mass m rotates, in the vertical plane (XOY), about the axis OZ. The position of the rod with respect to R is indicated by the angle θ (See Figure below).

Determine the kinetic elements of the solid, at point O, with respect to frame R.

Correction

Given:

- Rotation about OZ: $\vec{\omega} = \dot{\theta}\vec{Z}$
- Homogeneous rod: length L, mass m
- Center of mass G at midpoint

1. Momentum $\vec{P}(S/R)$:

$$\vec{P}(S/R) = m\vec{V}(G/R)$$

Distance from O to G: $OG = \frac{L}{2}$
Velocity of G: $V(G/R) = \frac{L}{2}\dot{\theta}$ (tangential direction)
Thus:

$$\vec{P}(S/R) = m \cdot \frac{L}{2}\dot{\theta}\ \vec{e}_\theta$$

2. Angular Momentum at O, $\vec{\sigma}(O; S/R)$:

$$\vec{\sigma}(O; S/R) = J(O, S) \cdot \vec{\omega}$$

For a thin rod rotating about one end (axis perpendicular to rod):

$$J(O, S) = \frac{1}{3}mL^2$$

Thus:

$$\vec{\sigma}(O; S/R) = \frac{1}{3}mL^2\dot{\theta}\ \vec{Z}$$

Kinetic Torsor at O:

$$\tau_C(S/R) = \left[\vec{P} = m\frac{L}{2}\dot{\theta}\ \vec{e}_\theta, \quad \vec{\sigma}(O) = \frac{1}{3}mL^2\dot{\theta}\ \vec{Z}\right]$$

Exercise 3:

A homogeneous and non-deformable hoop of radius R, center G, and mass m, rolls without slipping on an inclined plane of angle α.

We denote by x the abscissa of G and by θ the angle that indicates the position of a point on the hoop. Initially, the geometric contact point coincides with point O and $\theta = 0$.

1. What is the number of degrees of freedom of the hoop?

2. Determine the kinetic elements of the hoop at point O.

Correction

1. Degrees of Freedom:
Rolling without slipping condition:

$$x = R\theta$$

This constraint reduces the independent coordinates from 2 (x and θ) to 1.
Therefore: **1 degree of freedom**
2. Kinetic Elements at O:

- **Angular velocity:** $\vec{\omega} = \dot{\theta}\vec{Z}$
- **Velocity of G:** $V(G/R) = R\dot{\theta}$ (from rolling condition)
- **Momentum:**
$$\vec{P}(S/R) = m\vec{V}(G/R) = mR\dot{\theta}\ \vec{X}$$

Angular Momentum at O:
Using Koenig's theorem II:

$$\vec{\sigma}(O; S/R) = J(G, S) \cdot \vec{\omega} + m \cdot \vec{OG} \wedge \vec{V}(G/R)$$

For a hoop about its center:

$$J(G, S) = mR^2$$

$$\vec{OG} = R\vec{Y} \quad \text{(assuming contact point } O \text{ is directly below } G)$$

$$\vec{V}(G/R) = R\dot{\theta}\vec{X}$$

$$\vec{OG} \wedge \vec{V}(G/R) = (R\vec{Y}) \wedge (R\dot{\theta}\vec{X}) = R^2\dot{\theta}(\vec{Y} \wedge \vec{X}) = -R^2\dot{\theta}\vec{Z}$$

Thus:

$$\vec{\sigma}(O; S/R) = mR^2\dot{\theta}\vec{Z} + m(-R^2\dot{\theta}\vec{Z}) = 0$$

Kinetic Torsor at O:

$$\tau_C(S/R) = \left[\vec{P} = mR\dot{\theta}\ \vec{X}, \quad \vec{\sigma}(O) = \vec{0}\right]$$

Exercise 4:

A homogeneous solid cone S, of mass m, radius R, and height h is in motion about its fixed apex O, relative to a reference frame $T_1(O, X_1, Y_1, Z_1)$. Let $T(O, X, Y, Z)$ be the frame attached to the solid such that the OZ axis is the axis of revolution symmetry for the solid.

1. What is the number of degrees of freedom of the solid?
2. Express, in the Resal basis, the rotation vector of S with respect to T_1.
3. Determine the kinetic elements of the solid at point O with respect to T_1.

Correction

1. Degrees of Freedom:
A cone rotating about its fixed apex O has:

- 3 rotational degrees of freedom (Euler angles)

Therefore: **3 degrees of freedom**
2. Rotation Vector in Resal Basis:
In the Resal basis $(\vec{X}, \vec{Y}, \vec{Z})$ attached to the solid:

$$\vec{\omega}_{S/T_1} = \omega_X \vec{X} + \omega_Y \vec{Y} + \omega_Z \vec{Z}$$

where $\omega_X, \omega_Y, \omega_Z$ are the components of the angular velocity in the body-fixed frame.
3. Kinetic Elements at O:
Since O is fixed:

- **Velocity of O:** $\vec{V}(O/T_1) = \vec{0}$
- **Momentum:**
$$\vec{P}(S/T_1) = m\vec{V}(G/T_1)$$
But $\vec{V}(G/T_1) = \vec{\omega} \wedge \vec{OG}$, and $OG = \frac{3}{4}h$ along OZ for a cone.
- **Angular Momentum at O:**
$$\vec{\sigma}(O; S/T_1) = J(O, S) \cdot \vec{\omega}_{S/T_1}$$

For a cone about its apex, the inertia matrix is diagonal in the body-fixed frame:

$$J(O, S) = \begin{pmatrix} A & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C \end{pmatrix}$$

where for a homogeneous cone:

$$A = \frac{3}{20}m(R^2 + 4h^2), \quad C = \frac{3}{10}mR^2$$

Thus:
$$\vec{\sigma}(O; S/T_1) = A\omega_X \vec{X} + A\omega_Y \vec{Y} + C\omega_Z \vec{Z}$$

Kinetic Torsor at O:

$$\tau_C(S/T_1) = \left[\vec{P} = m\vec{\omega} \wedge \vec{OG}, \quad \vec{\sigma}(O) = A\omega_X \vec{X} + A\omega_Y \vec{Y} + C\omega_Z \vec{Z}\right]$$

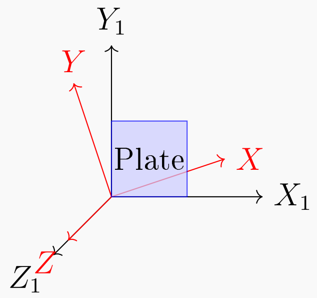
Exercise 5:

A homogeneous and non-deformable square plate $OABC$, of side a and mass m, is in motion about point O, origin of a reference frame $T_1(O, X_1, Y_1, Z_1)$, such that one of its sides OA remains in contact with the plane OX_1Y_1.

1. Represent, on a figure, the frame $T(O, X, Y, Z)$ attached to the solid and identify the Euler angles associated with this solid.
2. What is the number of degrees of freedom of the system?
3. Determine the inertia operator J_O of the solid at point O.
4. Express, in the basis of T, the rotation vector of the solid with respect to T_1.
5. Express, in the basis of T, the kinetic torsor of the solid, at O, with respect to T_1.
6. Determine the kinetic energy of the solid with respect to T_1.

Correction

1. Frame and Euler Angles:



Euler angles:

- ψ: Precession angle (rotation about Z_1)
- θ: Nutation angle (tilt of the plate)
- ϕ: Proper rotation (rotation about body Z-axis)

2. Degrees of Freedom:
Constraint: Side OA remains in contact with plane OX_1Y_1
This imposes 1 constraint, so from 3 rotational DOF we get: **2 degrees of freedom**
3. Inertia Operator J_O:
For a square plate of side a and mass m about one corner:

$$J_O = \begin{pmatrix} \frac{1}{3}ma^2 & 0 & 0 \\ 0 & \frac{1}{3}ma^2 & 0 \\ 0 & 0 & \frac{2}{3}ma^2 \end{pmatrix}$$

4. Rotation Vector in Body Frame:

$$\vec{\omega}_{S/T_1} = \omega_X \vec{X} + \omega_Y \vec{Y} + \omega_Z \vec{Z}$$

In terms of Euler angles:

$$\omega_X = \dot{\phi} \sin\theta \sin\psi + \dot{\theta} \cos\psi$$

$$\omega_Y = \dot{\phi} \sin\theta \cos\psi - \dot{\theta} \sin\psi$$

$$\omega_Z = \dot{\phi} \cos\theta + \dot{\psi}$$

5. Kinetic Torsor at O:
Since O is fixed:

- $\vec{V}(O/T_1) = \vec{0}$
- $\vec{P}(S/T_1) = m\vec{V}(G/T_1) = m(\vec{\omega} \wedge \vec{OG})$
- $\vec{\sigma}(O; S/T_1) = J_O \cdot \vec{\omega}_{S/T_1}$

Thus:

$$\tau_C(S/T_1) = \left[\vec{P} = m(\vec{\omega} \wedge \vec{OG}), \quad \vec{\sigma}(O) = \frac{1}{3}ma^2(\omega_X \vec{X} + \omega_Y \vec{Y}) + \frac{2}{3}ma^2\omega_Z \vec{Z} \right]$$

6. Kinetic Energy:
Using the formula for kinetic energy with fixed point O:

$$E_C(S/T_1) = \frac{1}{2}\vec{\omega}_{S/T_1} \cdot \vec{\sigma}(O; S/T_1)$$

$$E_C(S/T_1) = \frac{1}{2} \begin{pmatrix} \omega_X & \omega_Y & \omega_Z \end{pmatrix} \begin{pmatrix} \frac{1}{3}ma^2 & 0 & 0 \\ 0 & \frac{1}{3}ma^2 & 0 \\ 0 & 0 & \frac{2}{3}ma^2 \end{pmatrix} \begin{pmatrix} \omega_X \\ \omega_Y \\ \omega_Z \end{pmatrix}$$

$$E_C(S/T_1) = \frac{1}{6}ma^2(\omega_X^2 + \omega_Y^2 + 2\omega_Z^2)$$